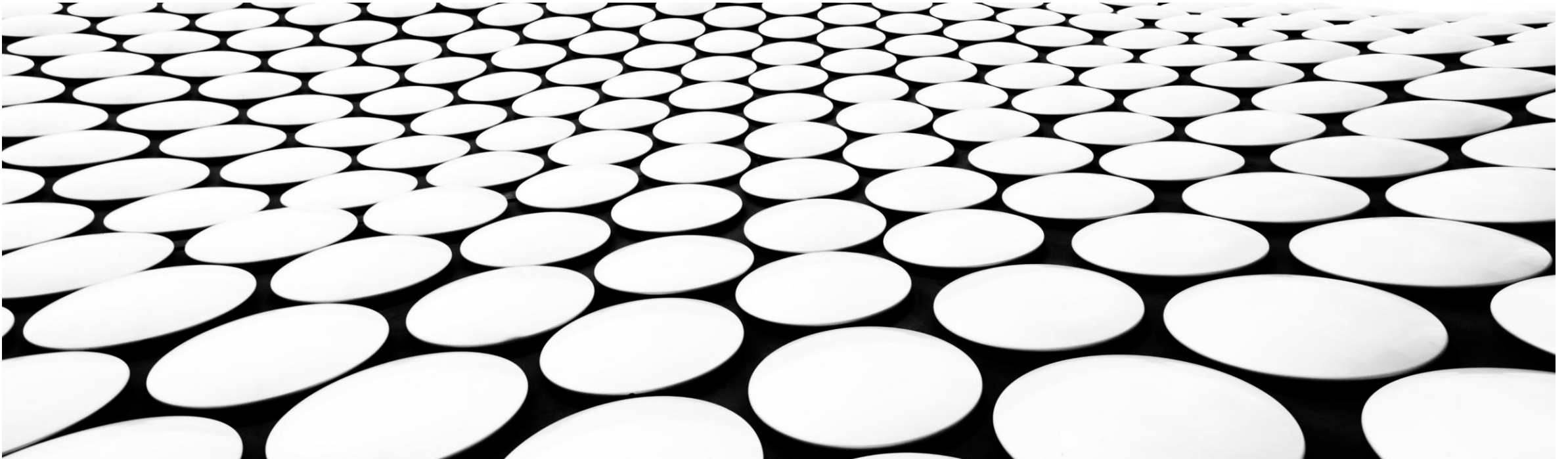
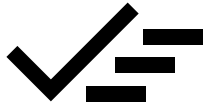


REVIEW SESSION





REVIEW SESSION

- Go through 25 example questions for the different weeks
- Point to important material throughout these questions
- Point to question styles throughout these questions

Week 1 - Insights from Data

Question 1

Many questions will be MCQs.

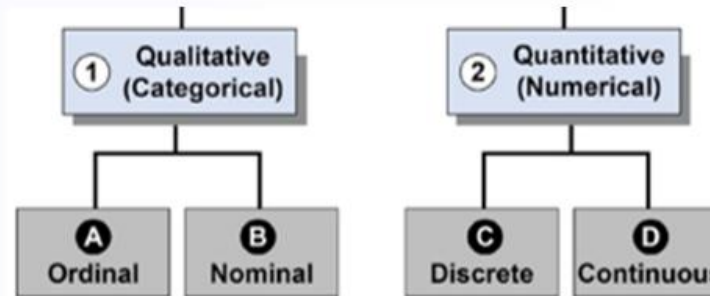
If an organization is trying to develop new policies for diabetes prevention, what type of analytics is in line with their goal?

- ☒ a) Prescriptive Analytics
- ☐ b) Descriptive Analytics
- ☐ c) Predictive Analytics
- ☐ d) Diagnostic Analytics

Question 2

Some questions use images/data from the slides.

Among the 4 types of data defined



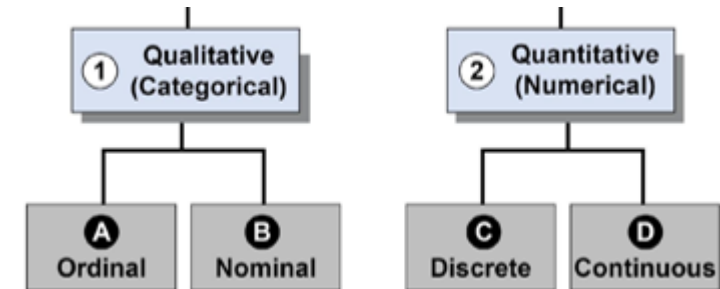
which one would be assigned, in the medical domain, to a feature "Treatment Type" that could be surgery, medication, etc.

- ☐ a) Categorical Ordinal
- ☒ b) Categorical Nominal
- ☐ c) Numerical Discrete
- ☐ d) Numerical Continuous

not numerical
no order

**Dataset
schema**

Column	Unit/Description
Patient ID	Unique identifier (alphanumeric)
Age	Years
Gender	Male/Female
Length of Stay	Days
Number of Diagnoses	Count
Primary Diagnosis	ICD Code
Comorbidities	Count of co-existing conditions
Treatment Type	Categorical (e.g., surgery, medication)
Readmission Within 30d	Yes/No



Question was inspired
from this slide.

Question 3

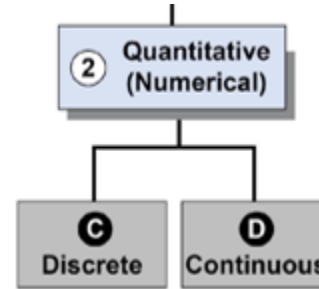
Some questions are Multi-Select Questions. Make sure you select all that applies.

In a medical domain, why would "Number of patients in a clinic" be a better example of discrete numerical than "Age of patients in a clinic"?

- ☒ a) Because Age is dividable and could be seen as a continuous numerical data.
- ☒ b) Because the number of people cannot be a fraction
- ☐ c) Because we can do an average of age and not an average of number of patients
- ☐ d) Because age has a smaller range (e.g. 0-120) than number of patients (e.g. 0-500)



- **Definition:** Data that can take only specific, distinct values, often counted. Values are usually whole numbers.
- **Examples:**
 - **Health:** Number of patients in a clinic, number of medications a patient takes
 - **Environment:** Number of trees in a park, number of rainy days in a month
 - **Business:** Number of items sold, number of employees in a company



- **Definition:** Data that can take any value within a given range and is often measured. Values can include decimals and fractions.
- **Examples:**
 - **Health:** Body temperature (e.g., 98.6°F), weight (e.g., 68.4 kg), blood pressure (e.g., 120.5 mmHg)
 - **Environment:** Air quality index (e.g., 56.7), rainfall (e.g., 12.8 cm), temperature (e.g., 21.3°C)
 - **Business:** Revenue (e.g., \$10,254.75), product dimensions (e.g., 12.45 cm)

Question 4

You put a survey out to obtain, for 1000 students, the type of food they would like to have on University of Ottawa campus. You wish to use such dataset for your study. How would you characterize this data along the dimensions presented in the lecture.

- ☒ a) Qualitative, Primary, Internal
- ☐ b) Qualitative, Structured, External
- ☐ c) Quantitative, Primary, Real-Time
- ☐ d) Qualitative, Semi-structured, Historical

Even if you're not entirely in agreement with all info... only one answer is possible.

Different Types of Data



Khushee Kapoor · [Follow](#)

5 min read · Aug 28, 2021

Measurement

Quantitative (Numerical)
Qualitative (Categorical)

Format

Structured
Unstructured
Semi-Structured

Volume

Big Data
Small Data

Collection

Primary Data
Secondary Data

Source

Internal Data
External Data

Time

Historical Data
Real-Time Data

Week 2 - Exploratory Data Analysis

Question 5

Some questions are True/False. Those are worth 0.5.

The range is always smaller than the mode.

- ☐ True
☒ False

Always – try to find a counterexample = False

If « sometimes » or « could be », find an example = True

Statistics Handbook for Data Analysts



Anita Gupta · Follow
17 min read · Sep 14, 2024

Statistics Essentials: Understanding
Key Terms in Data Analysis

Measures of central tendency

1. Mean
2. Median
3. Mode

Measures of dispersion

1. Range
2. Variance
3. Standard deviation

Question 6

Assuming the ONLY data you have about the Census is the one shown in the table below.

	age	work-class	fnlwgt	education	education-num	marital-status	occupation	relationship	race	sex
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	White	Male
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	White	Male
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	White	Male
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband	Black	Male
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife	Black	Female

What would be the average number of years of education of people?

- ☒ a) 11
- ☐ b) 13
- ☐ c) 10,5
- ☐ d) 11,2

A few questions will be applicative (using some formulas).

Question 7

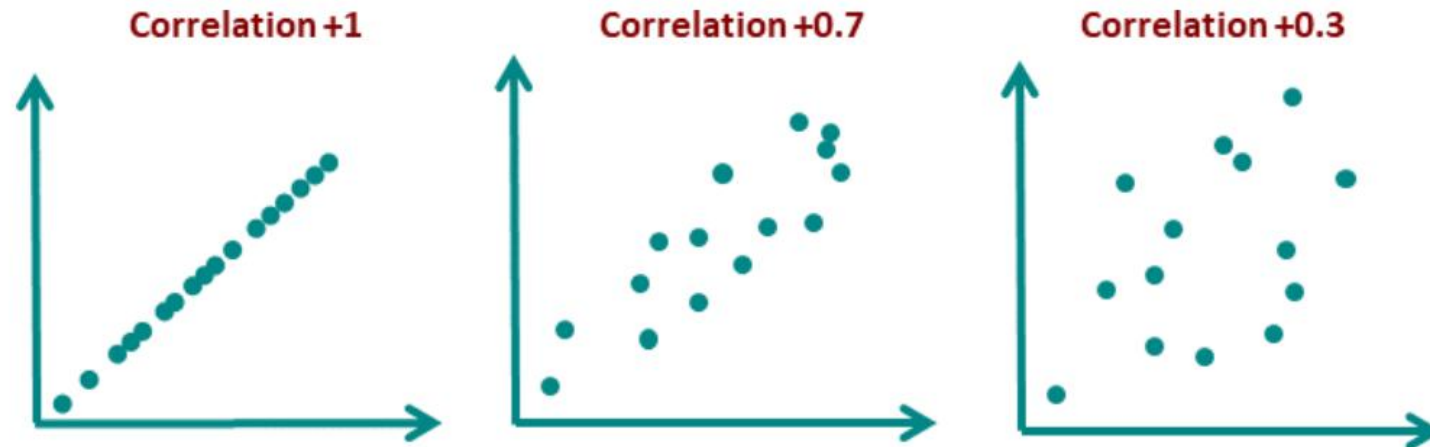
Pearson correlation varies from -1 to +1.

- ☒ True
- ☐ False



Pay attention to the slide information.
Strength is absolute value. But sign
(pos/neg) indicates direction.

$ r $	Strength of correlation
$0.0 < 0.1$	no correlation
$0.1 < 0.3$	little correlation
$0.3 < 0.5$	medium correlation
$0.5 < 0.7$	high correlation
$0.7 < 1$	very high correlation



Question 8

Scatter plots are to show the relation between two categorical variables.

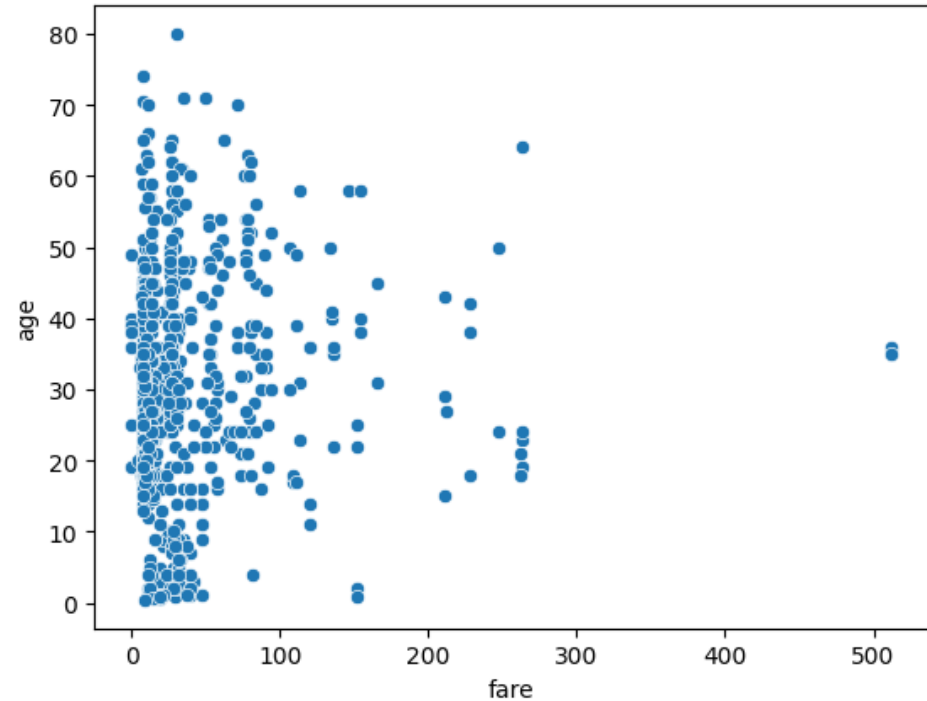
- ☐ True
- ☒ False

Review types of plots for different
bivariate and univariate analysis

Scatter Plots

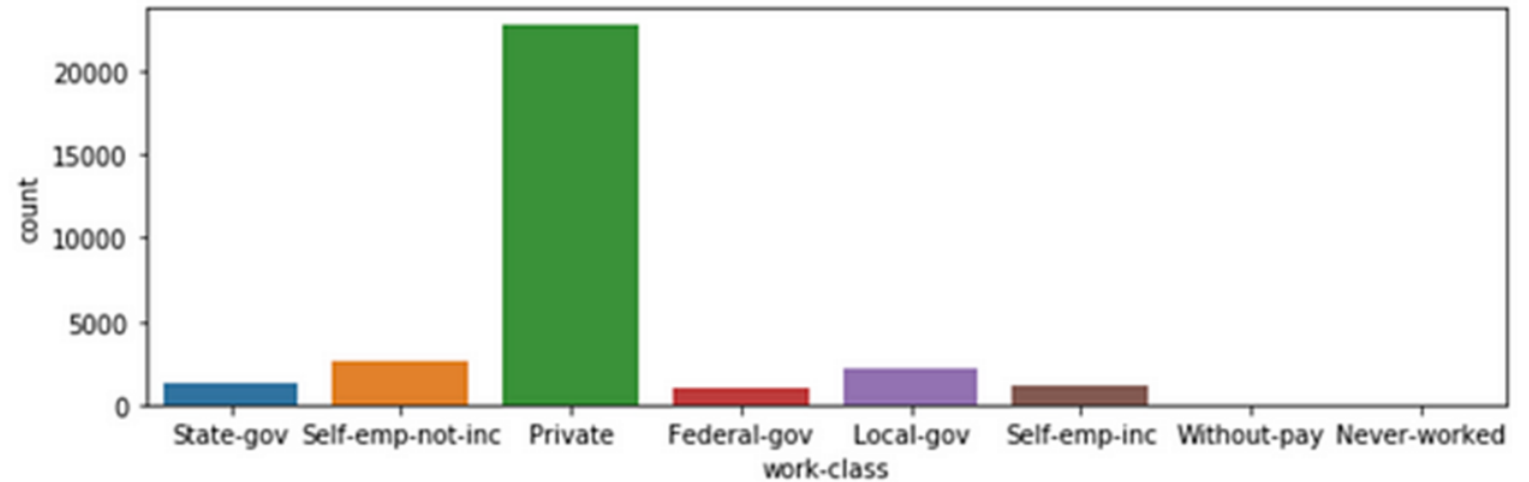
A **scatter plot** shows the relationship between two continuous variables, making it great for visualizing correlation.

```
sns.scatterplot(x="fare", y="age", data=titanic)
```



Question 9

The countplot shown below to present the categorical feature of work-class from the Census Data does not provide a good data visualization tool, mainly because of :



- ☒ a) A large difference between the frequency of the mode and the frequency of other values.
- ☐ b) Two values that obtain a similar count.
- ☐ c) Values with near-zero counts.
- ☐ d) A set of colours that might not please the readers.

Week 3 - Data Collection

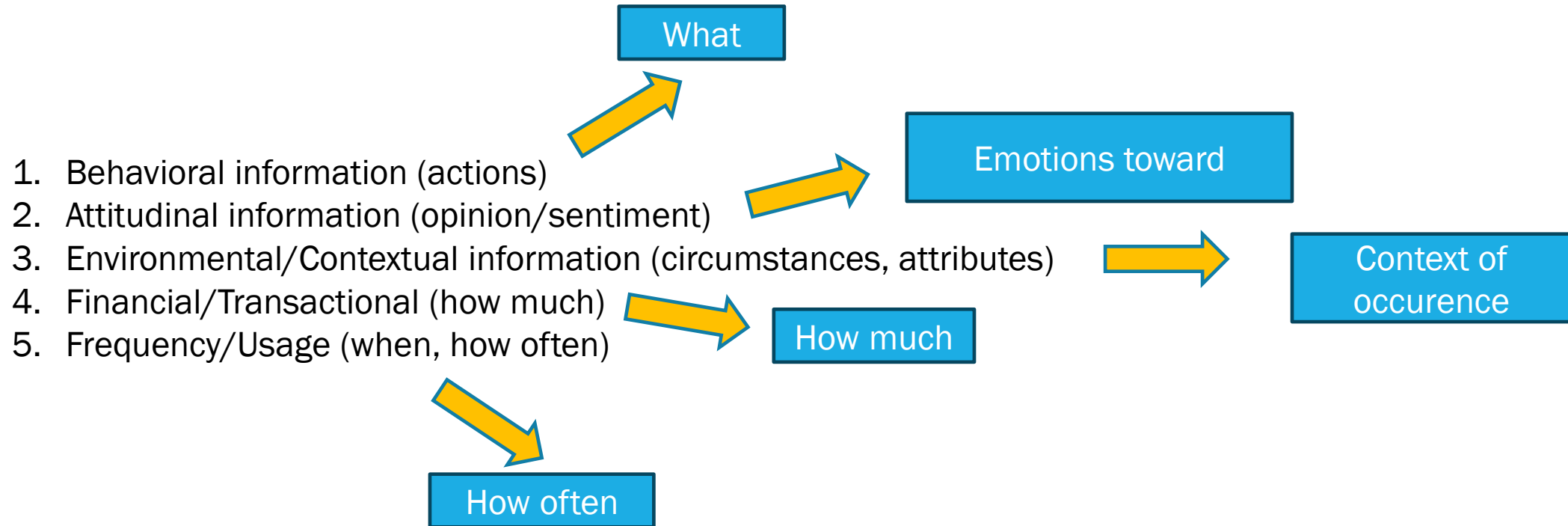
Question 10

In Data Collection, there are different elements we try to capture.

If the insight we wish to get at is : *How do air quality levels vary across different regions of Ottawa during peak traffic hours?*

How would we characterize the "variation of air quality level" part?

- ☒ a) Behavioral Information
- ☐ b) Attitudinal Information
- ☐ c) Contextual Information
- ☐ d) Transactional Information



Question 11

Among the following self-reported data collection methods, which one assume that participants will discuss with each other?

- ☒ a) Focus groups
- ☐ b) Surveys
- ☐ c) Interviews
- ☐ d) None of the other 3 choices provided

Self-Reported Data	Surveys Interviews Focus groups
Data directly provided by individuals	
Behavioral and Observational Data	Observations Experiments
Data collected through monitoring or observing human behavior.	

Automated Data	Logs Sensor Transactions
Data generated obtained through an automated system.	
Digital and Platform-Based Data	Social Media Data Web Scraping APIs
Data originating from digital platforms, systems, or online activity.	

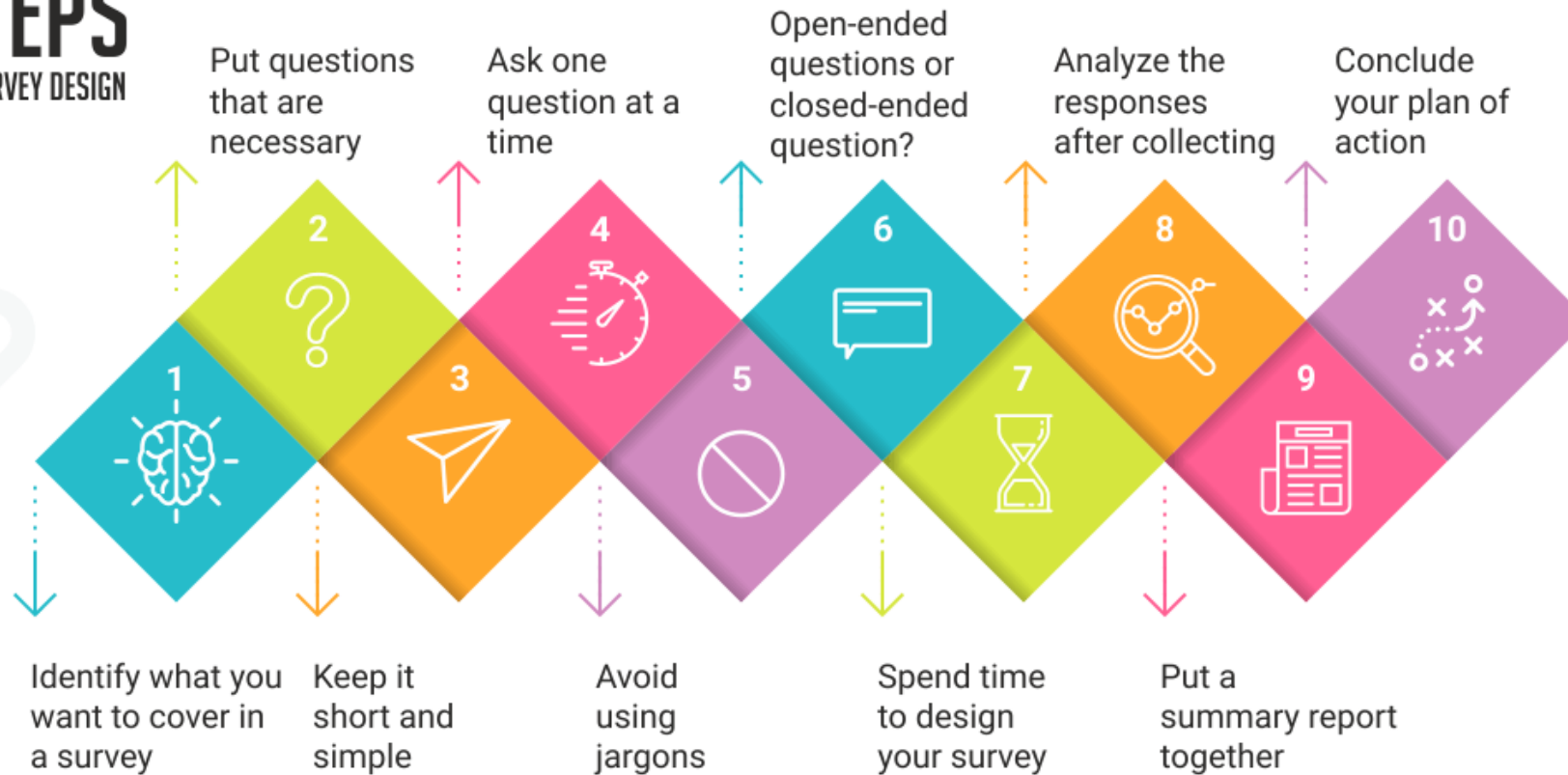
Question 12

Some are easy questions, but you need to know where to find the answer.

Which step is not included in the Survey Design Steps presented in class (as suggested by QuestionPro) ?

- ☐ a) Avoid using jargons
- ☐ b) Keep it short and simple
- ☐ c) Analyze the response after collecting
- ☒ d) Make all questions objective

10 STEPS TO A GOOD SURVEY DESIGN



Question 13

Among the data collection methods presented, for which did I mention that it was becoming so popular that a whole offering of analysis platforms was being developed?

- ☒ a) Log analysis
- ☒ b) Social Media analysis
- ☒ c) Web scraping analysis
- ☐ d) API analysis

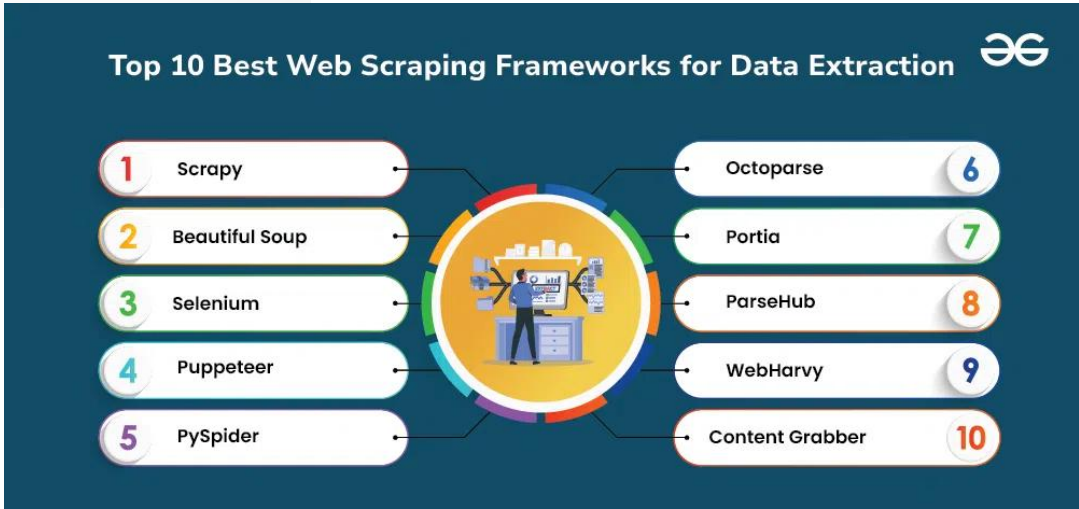
12 Best Social Media Analytics Tools in 2026

Here's a summary of the top social media analytics tools we'll be reviewing:

1. Whatagraph
2. Metrics Watch
3. Keyhole
4. Google Analytics 4 (GA4)
5. Sprout Social
6. Buffer
7. Hootsuite
8. Brandwatch
9. Rival IQ
10. Zoho Social
11. Mentionlytics
12. BuzzSumo

Best Log Analyzers

1. **Sematext Logs**
2. Elastic Stack
3. Graylog
5. Loggly
5. Splunk
6. Logz.io
7. Sumo Logic
8. SolarWinds
9. ManageEngine
10. Papertrail
11. Mezmio
12. Datadog
13. LogicMonitor



Question 14

Assume survey respondents responded that their favorite shirt color is grey, but then in-store observations showed that most customers went to the rack with green shirts, and purchase information showed that the blue shirts are running low on stock. What needs to be considered when seeing these results?

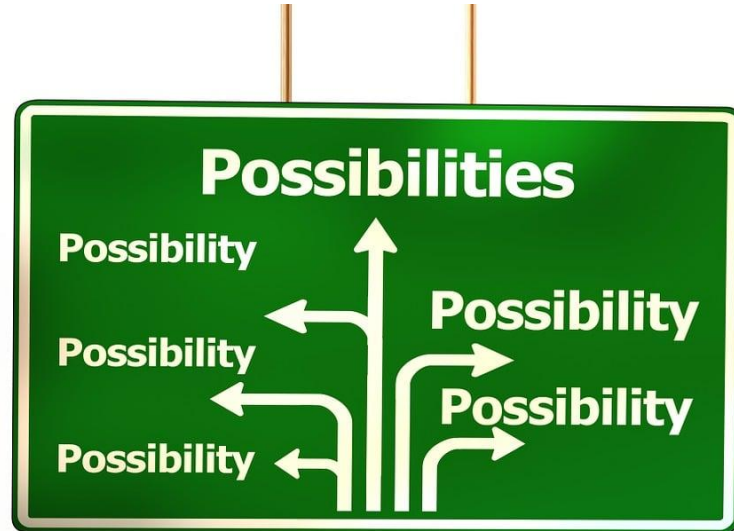
- ☒ a) Self-reported data is based on memory and perception.
- ☒ b) Transactional data is objective.
- ☐ c) Observation is subjective.
- ☐ d) The number of customers studied for all three methods should be the same to make comparisons.

Question 15

A Data Collection Decision Factor is Reliability... nevertheless, all methods can suffer from reliability problems. Highlight what would be considered a reliability problem reflected in the data.

- ☒ a) In surveys, participants misunderstand questions because the questions are not clear.
- ☒ b) In an A/B testing, all A version sites are shown in the morning and all B version sites are shown in the afternoon.
- ☐ c) In traffic Observation, no car arrive at a particular intersection all day.
- ☒ d) In sensor data gathering, multiple temperature sensors at the same position do not register the same information.

Data Collection



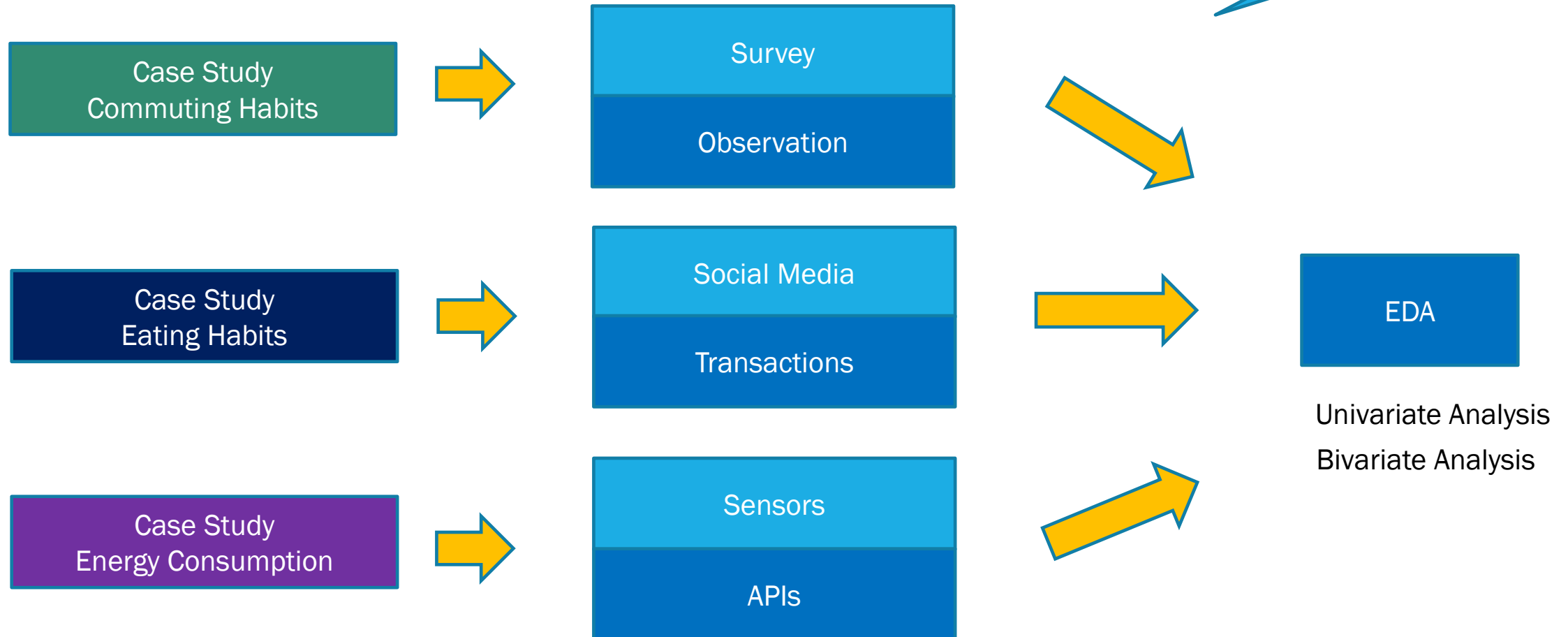
1. Purpose (Problem Statement/Question/Goal of Study)
2. Resources (Time/Effort/Cost)
3. Availability of already collected data
4. Domain Expertise
5. Technical Expertise
6. Data Storage possibilities (cost)
7. Reliability
8. Ease of use (feasibility) and integration
9. Ethical and Privacy concerns



Case studies:

Focus on two collection methods.

Review also... even if
no questions here



Week 4 - Data Cleaning

(data quality, duplicates, validation)

Question 16

Provided the following situation:

We devised a survey on food consumption and exercising, but did not include a question on the number of hours of exercise per day.

Which of these 4 data quality dimensions does it concern?

- ☒ a) Completeness
- ☐ b) Accuracy
- ☐ c) Consistency
- ☐ d) Validity



The 6 data quality dimensions with examples

Data quality goes beyond what cleaning can do...

1. Completeness

How complete is your dataset?

2. Accuracy

Is the data an accurate representation of the element it describes?

3. Consistency

Is your data synchronized across your organization?

4. Validity

Does the data fit the defined range and definition?

5. Uniqueness

Are there any duplicates within your data?

6. Integrity

Can your data be traced and connected across the organization?

Question 17

Provided the following situation:

Unfortunately, at input time, users could enter their preferred color by typing the name of the color, so there are spelling errors needed to be corrected.

Which of these 4 data quality dimensions does it concern?

- ☐ a) Completeness
- ☐ b) Accuracy
- ☐ c) Consistency
- ☒ d) Validity

Question 18

Within the cleaning task of duplicate removal, if the algorithm removes a row that is not an actual duplicate, it increases the number of False Positives.

☒ True

☐ False

Review exact duplicates and near-duplicates and what can cause false positives and false negatives

Question 19/20

Review these errors

- 1) Data Type Check
- 2) Code Check
- 3) Range Check
- 4) Format Check
- 5) Consistency Check
- 6) Uniqueness Check
- 7) Presence Check
- 8) Length Check
- 9) Look Up

Verifying that a number of items bought column contains only numbers is an example of "length check".

- ☐ True
- ☒ False

Verifying that a discounted food item price is less than the normal food item price is an example of "range check".

- ☐ True
- ☒ False

Week 4 - Data Cleaning

(Outlier Detection)

Question 21

In the example for IQR shown in the slides, we have:

Data
4
5
7
7
8
9
10
12
17
18
19
22
44

Assume the data entry "22" in the example shown is replaced by "34", then the "44" previously considered as an outlier would not be an outlier anymore as it would be closer to the second largest value 34.

☐ True

☒ False

Data
4
5
7
7
8
9
10
12
17
18
19
22
44



$Q1 = 7$



Median = 10



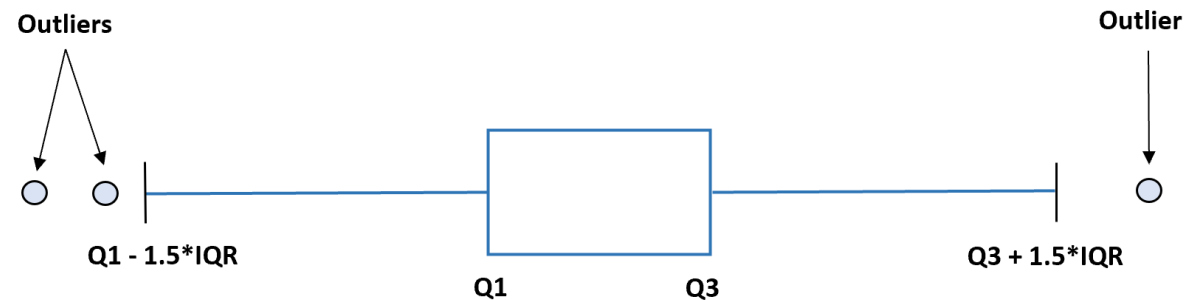
$Q3 = 18.5$

$$IQR = 18.5 - 7 = 11.5$$

$$Q1 - 1.5 * IQR \leq \text{Acceptable range} \leq Q3 + 1.5 * IQR$$

$$Q1 - 17.25 \leq \text{Acceptable range} \leq Q3 + 17.25$$

$$-10.25 \leq \text{Acceptable range} \leq 35.75$$



Z-score approach

Question 22

If using z-score for outlier detection, a threshold is set in relation to:

- ☒ a) Number of std from the mean found acceptable
- ☐ b) Number of outliers anticipated
- ☐ c) Average of the distribution
- ☐ d) Number of datapoints in the dataset

Formula

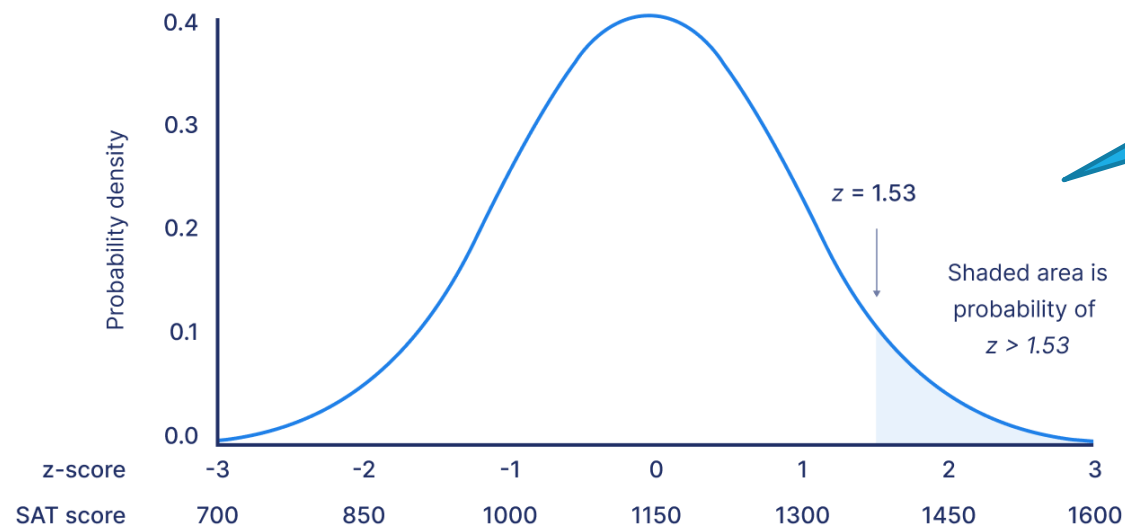
Calculation

$$z = \frac{x - \mu}{\sigma}$$

$$z = \frac{1380 - 1150}{150}$$

$$z = 1.53$$

Standard normal distribution

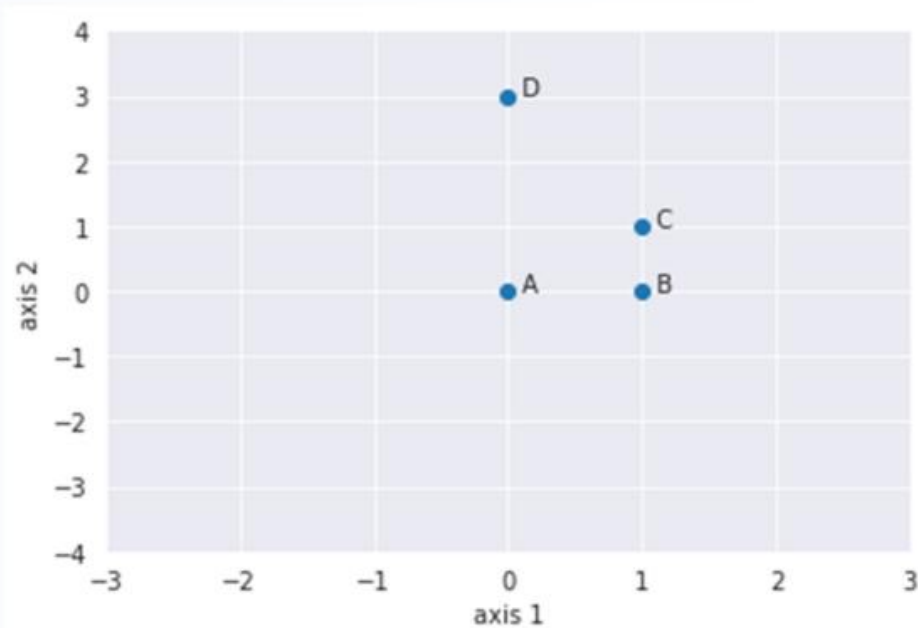


Then z-score obtained can be thought as number of std away from the mean

kNN approach

Question 23

Assume the following data. Assume Manhattan Distance is used.



We are interested in using a kNN outlier detection approach. If k is set to 1, what is $kDistance(D)$?

- ☒ a) 3
- ☐ b) 2
- ☐ c) 1
- ☐ d) 4



Manhattan_Distance(A,B) =	1
Manhattan_Distance(A,C) =	2
Manhattan_Distance(A,D) =	3
Manhattan_Distance(B,C) =	1
Manhattan_Distance(B,D) =	4
Manhattan_Distance(C,D) =	3

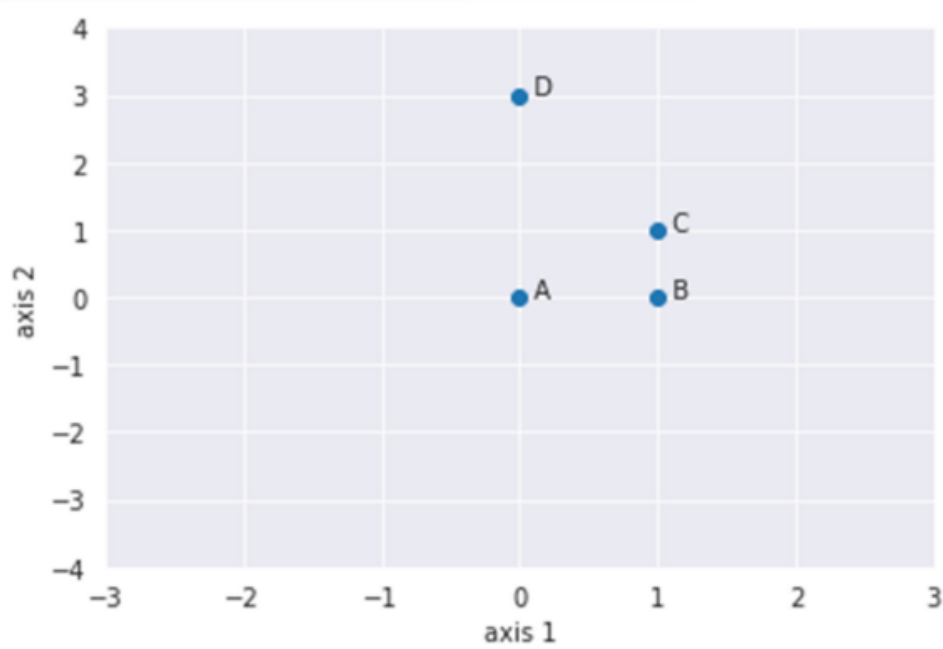
Review different distances
presented

Review the notion of
kDistance

Question 24

LOF approach

Let's focus on the LOF approach used with the following dataset:



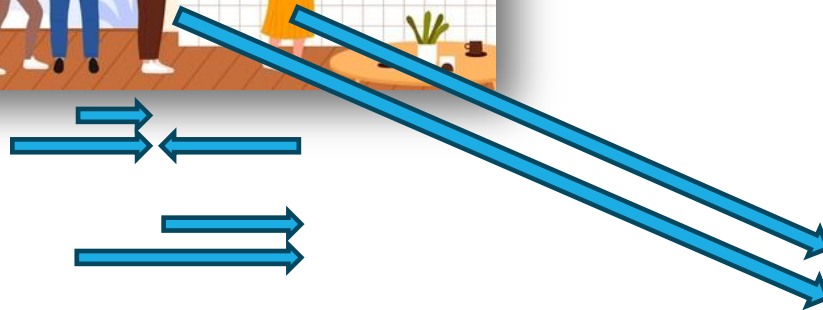
Assume k is 1. Distance used is Manhattan.

What would be the neighborhood of A and B ?

- ☒ a) $N_1(A) = \{B\}$ and $N_1(B) = \{A, C\}$
- ☐ b) $N_1(A) = \{B, C\}$ and $N_1(B) = \{A, C\}$
- ☐ c) $N_1(A) = \{C\}$ and $N_1(B) = \{A, C, D\}$
- ☐ d) $N_1(A) = \{B, C\}$ and $N_1(B) = \{C, D\}$



- We need a Distance Measure, and set K (e.g. 2).
- We need a notion of Neighbors (e.g. 2)
- We need a notion of Reachability
- We need a notion of Average Reachability from the neighbors
- We need to compare that average reachability to that of the neighbors



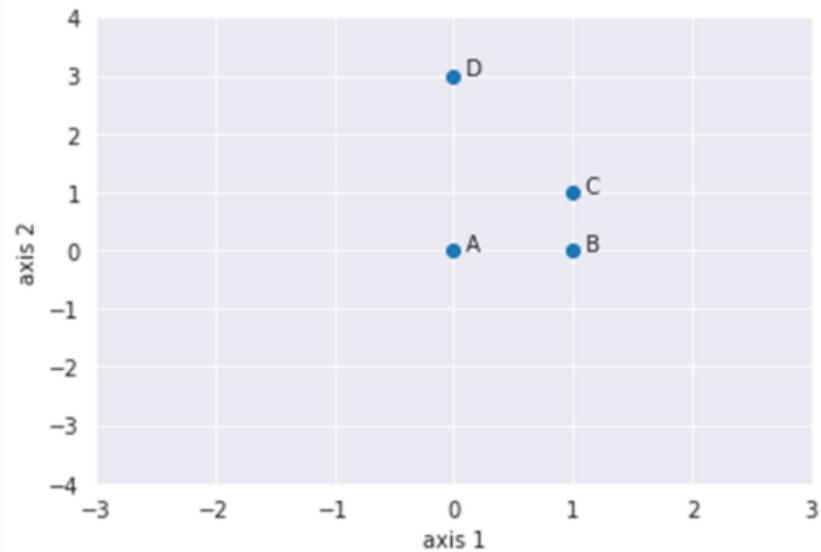
Reachability from
its neighbors

Can I be reached by my neighbors the same way that my neighbors can be reached by their neighbors.?

LOF approach

Question 25

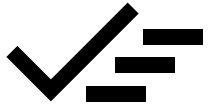
Let's focus on the LOF approach used with the following dataset:



Assume k is 1. Distance used is Manhattan.

What would be the Local Reachability Density for D, so $LRD_1(D)$?

- ☐ a) $LRD_1(D) = 1/6$
- ☒ b) $LRD_1(D) = 1/3$
- ☐ c) $LRD_1(D) = 1/2$
- ☐ d) $LRD_1(D) = 1$



REVIEW SESSION

- Go through 25 example questions for the different weeks
- Point to important material throughout these questions
- Point to question styles throughout these questions